

Aaron Daniel **KARTMANN**



PERSONAL PROFILE TEXT

I am an aspiring experimental condensed matter physicist with a strong interest in technological applications. Motivated by the challenges of the climate crisis, I aim to contribute to the development of new technologies that support a more sustainable future. I am experienced in laser experiments and electronic transport under high magnetic fields as well as strain.

CONTACT DETAILS

 [linkedin.com/in/aaronkartmann](https://www.linkedin.com/in/aaronkartmann)
 github.com/aaronkartmann
 aaronkartmann.github.io

SKILLS

- High Magnetic Fields, Electronic Transport, Torque Magnetometry, Uniaxial Strain
- (Pulsed) Lasers, Optics, Faraday Spectroscopy, Helicity-Dependent All-Optical Switching
- Cryogenics (N, He-4, He-3)
- Python: Numpy, Data Analysis, (Numerical) Simulations, Machine Learning
- C++: Classes, Computationally Efficient Programming, Python Integration
- Cutting and contacting microscopic samples
- Electrical contact fabrication via evaporation
- MS Office, Git, LaTeX, Linux, OriginLab
- Teaching, Presenting, Problem Solving
- Scientific Writing

EXPERIENCE

MASTER INTERNSHIP at *HFML-FELIX Nijmegen*.

09.2025–08.2026

- ◇ At the Correlated Electron Systems group under Steffen Wiedmann.
- ◇ Investigated magnetoresistance, superconductivity, and Fermi surface of topological materials in magnetic fields using electronic transport and torque magnetometry, including experiments under applied uniaxial strain.
- ◇ Gained laboratory experience with superconducting and Bitter magnets, cryostat operation, vacuum equipment, and sample preparation.

TEACHING ASSISTANT at *Radboud University Nijmegen*.

09.2023–11.2023, 09.2024–11.2024

- ◇ A teaching assistant leads tutorials, answers student questions, provides guidance on exercises, and grades assignments.
- ◇ Gained experience in teaching, leading groups, and explaining complex problems in simple terms.
- ◇ Assisted in teaching the first-semester course Warmteleer (Introduction to Thermodynamics) during the academic years 2023/24 and 2024/25.

BACHELOR INTERNSHIP at *Radboud University Nijmegen*.

02.2024–07.2024

- ◇ At the Spectroscopy of Solids and Interfaces group under Theo Rasing.
- ◇ Investigated helicity-dependent all-optical switching of magnetization induced by picosecond laser pulses in Pt/Co/Pt thin films.
- ◇ Gained laboratory experience in optics and working with lasers, specifically Faraday microscopy and magneto-optical Faraday measurements.
- ◇ Possible applications in hard disk drives for data centers.

EDUCATION

MASTER OF SCIENCE IN PHYSICS, SPECIALIZATION QUANTUM MATTER.
Radboud University Nijmegen. **2024–2026**

Final GPA: 8.8/10 (Cum Laude)

BACHELOR OF SCIENCE IN PHYSICS AND ASTRONOMY.
Radboud University Nijmegen. **2021–2024**

Final GPA: 8.3/10 (Cum Laude)

LANGUAGES

- ◇ GERMAN, C2.
- ◇ ENGLISH, C1.
C1 Advanced Cambridge Certificate. **06.2021**
- ◇ DUTCH, C1.
RU-NT2-exam on level B2. **08.2021**
- ◇ FRENCH, A1.